Chunxiao (Calvin) Lin

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Research Summary

FPGA researcher building **Echo State Network (reservoir computing) accelerators with on-chip learning** for **Next-G wireless**, from DSP-level microarchitecture to **real-time PC–FPGA testbeds** on live distributed-MIMO links. 12 publications, including first-author IEEE TCAD, ICCAD, ICNC, ISQED and Frontiers.

Education

Virginia Tech, Ph.D. in Computer Engineering — GPA: 4.0/4.0 — *Sep 2021 – May 2027 (Expected)*

- Advisor: Prof. Yang (Cindy) Yi | BRICCS@VT (Brain-Inspired Computing, Communication, and Security Lab)

University of Southern California, Master of Engineering — GPA: 3.95/4.0 — *Sep 2019 – May 2021*

- Focus: Computer Architecture, VLSI Design, Digital System Design

Technical Skills

Hardware: Verilog/SystemVerilog RTL, DSP48 datapaths, systolic arrays, fixed-point design; Xilinx Virtex-7, Zynq-7000, Zynq UltraScale+, RFSoC (ZCU216)

Systems: 10G/1G Ethernet (MAC, PCS/PMA), UDP/IP (lwIP), AXI, PS–PL co-design; USRP SDRs, GNU Radio, Keysight PROPSIM

Tools: Vivado, Vitis, Quartus, Cadence, ns-3; Python, MATLAB, C

Research Experience

Graduate Research Assistant, BRICCS@VT, Virginia Tech — *Sep 2021 – Present*

Real-Time PC–FPGA Testbeds for Distributed MIMO — *2025 – 2026*

- Built 1 GbE (PS–PL hybrid, lwIP) and 10 GbE (pure-RTL Ethernet) testbeds that close a live 4×2 DMIMO loop through USRPs and a PROPSIM F16 emulator, with one ESN predictor per user on the FPGA.
- **≤0.9 ms end to end**, **1,250 predictions/s per user** at 3.42 W; NMSE 0.0119 → 0.0087 versus outdated CSI.

On-chip ESN Learning and High-Speed Inference — *IEEE TCAD 2026, ICNC 2026* — *2024 – 2025*

- First complete on-chip ESN with RLS training on FPGA (systolic DSP48 readout, inversion-free pipelined RLS): **3.87 μs per training step**, **>1600× / 290× energy efficiency** vs. CPU / GPU, 5.3× lower NMSE than the prior on-chip FPGA learner.
- Hybrid DSP–CLB neuron with range-addressable tanh: **+25% DSP efficiency**, **65.98 MS/s**, 5.7–14.2 dB better NMSE than AR prediction; also ported to a Zynq PS–PL co-design.

ESN Symbol Detection for MIMO-OFDM — *ICCAD 2023, Frontiers 2024, ISQED 2022* — *2021 – 2023*

- Nine-DSP48 neuron and SDR/FPGA platform for over-the-air 2×2 MIMO-OFDM: **3.3× throughput** of the prior design, BER within 0.03% of software and below MMSE; scales to 4×64 massive MIMO.
- Bit-serial, CSD-encoded neuron hard-wiring the sparse weights: **80.7% fewer LUTs**, DSPs 162 → 16.

Other Projects

VLSI and Hardware Security (USC) — *DFT 2021* — *2020*

- 64×64 10T-SRAM for in-memory computing with leakage reduction; dynamically secured scan chain and the GF-Flush attack (500-bit keys in <7 s, >4300× faster than SAT).

PUBLICATIONS

- **C. Lin**, U. S. Khan, M. Zheng, M. F. Azmine, L. Liu, D. J. Jakubisin, Y. Yi, "FPGA-based Reservoir Computing with On-chip Learning for MIMO Channel Prediction," *IEEE TCAD*, 2026.
- S. S. Chowdhury, M. R. Sarkar, J. Ling, **C. Lin**, Y. Yi, "SPARC: Spiking Plasticity-Adaptive Reservoir Computing with On-Chip Reward-Modulated Triplet STDP for Spectrum Sensing," *IEEE TVLSI*, 2026 (accepted).
- **C. Lin**, U. S. Khan, L. Liu, Y. Yi, "FPGA-based High-speed Reservoir Computing for MIMO Channel Prediction," *ICNC*, 2026.
- R. Gaurav, S. Sinha, **C. Lin**, T. C. Stewart, L. Liu, Y. Yi, "Multivariate Legendre-SNN on Loihi-2 for Time Series Classification and 5G Jamming Detection," *IEEE TCASAI*, 2026.
- A. Liu, M. F. Azmine, **C. Lin**, Y. Yi, "Reservoir Computing: Foundations, Advances, and Challenges Toward Neuromorphic Intelligence," *AI (MDPI)*, 2026.
- A. Liu, **C. Lin**, D. Rosen, Y. Yi, "Echo State Networks in Reservoir Computing: Foundations, Benchmarks, and Applications to Next-G Wireless Communication," *IEEE ICDMW*, 2025.
- **C. Lin**, Y. Yi, "Enhancing FPGA CAD Flow with AI-Powered Solutions," *Springer book chapter*, 2025.
- K. Yang, Y. Hu, D. Chen, **C. Lin**, Y. Yi, P. A. Beerel, P. Nuzzo, "Machine Learning-Enhanced Analysis and Design for Trustworthy Integrated Circuits," *Springer book chapter*, 2025.
- **C. Lin**, M. F. Azmine, Y. Liang, Y. Yi, "Leveraging Neuro-Inspired AI Accelerator for High-Speed Computing in 6G Networks," *Frontiers in Computational Neuroscience*, 2024.
- **C. Lin**, M. F. Azmine, Y. Yi, "Accelerating Next-G Wireless Communications with FPGA-Based AI Accelerators," *IEEE/ACM ICCAD*, 2023.
- **C. Lin**, Y. Liang, Y. Yi, "FPGA-based Reservoir Computing with Optimized Reservoir Node Architecture," *ISQED*, 2022.
- D. Chen, **C. Lin**, P. A. Beerel, "GF-Flush: A GF(2) Algebraic Attack on Dynamically Secured Scan Chains," *IEEE DFT*, 2021.